Powers and Exponents Flagship

Lesson 6-1-flagship

Name: ______________________ Date: ______________________ Class: ______________________

SOUND STUDIO MISSION

The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In 2^3, the small 3 means multiply 2 by itself 3 times: $2 \times 2 \times 2 = 8$

Exponent

A small number that tells how many times to multiply the number by itself.

In 5^2, the base is 5 — it is the number being multiplied: $5 \times 5 = 25$

Base

The number that gets multiplied by itself.

$10^3 = 10 \times 10 \times 10 = 1{,}000$ — read as '10 to the third power' or '10 cubed'

Power

A number written with a base and an exponent, like 2^3.

Evaluate 3^4: write $3 \times 3 \times 3 \times 3$, then multiply step by step: $9 \times 9 = 81$

Evaluate

To find the value of an expression.

In 3x, the 3 is the coefficient — if $x = 4$, then $3x = 3 \times 4 = 12$

Coefficient

The number in front of a letter, like the 3 in 3x.

4x and 2x are like terms (both have x); 4x and $4x^2$ are NOT like terms (different powers)

Like terms

Terms with the same letter, like 2x and 5x.

Key Ideas & Notes

- You're a sound engineer at a music studio.
- Each time you flip a switch, the volume doubles.
- The starting volume is 1 unit.
- After flipping the switch 3 times, the volume is $2^3 = 2 \times 2 \times 2 = 8$ units — that's 8 times louder!
- Evaluate each exponential expression. Write the repeated multiplication and find the value.

Think About It

- What happens to the volume each time you flip a switch?
- Why is 2^3 equal to 8, not 6?
- How is repeated multiplication different from repeated addition?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Each switch flip doubles the volume: 1, then 2, then 4, then 8. How is each number related to the one before, and how could you write 2 times 2 times 2 a shorter way?

Level 1 support

Start here: Each step multiplies by 2 again, so repeated multiplication can be written as a power like 2 cubed.

Each number is the one before it times ____, so it is repeated ____.

Cada número es el anterior por ____, así que es ____ repetida.

2 times 2 times 2 can be written as 2 to the ____ power.

2 por 2 por 2 se puede escribir como 2 a la ____ potencia.

Word bank: base | exponent | power | evaluate | double

Listen for: *Students see each volume is the previous one times 2 and write $2 \times 2 \times 2$ as 2 cubed, naming 2 as the base and 3 as the exponent.*

Level 2

Predict the volume after 6 switch flips and write it as a power. Explain how the exponent connects to the number of flips.

- After 6 flips the volume is ____, which I write as ____.
- The exponent matches the number of flips because ____.

EXPLORE

In the power 2 cubed, which number is the base and which is the exponent? What job does each one do when you evaluate it?

Level 1 support

Start here: The base is the factor being multiplied; the exponent tells how many times to use it.

The base is ___ and it is the number being ___.

La base es ___ y es el número que se está ___.

The exponent is ___ and it tells how many ___.

El exponente es ___ e indica cuántas ___.

Word bank: **base** **exponent** **power** **evaluate** **factor**

Listen for: *Students identify the base as the repeated factor and the exponent as the count of factors, and evaluate 2 cubed as 2 × 2 × 2 = 8.*

Level 2

Why is 2 cubed different from 2 times 3? Evaluate both and explain what makes the answers different.

- 2 cubed equals ___ but 2 times 3 equals ___ because ___.
- An exponent means ___, while multiplication means ___.

CONNECT

When you bounce a track, the file size grows as you double the layers. Where else do you see quantities that grow by repeated multiplication?

Level 1 support

Start here: Doubling money, folding paper, or sharing a rumor all grow by repeated multiplication.

I see repeated multiplication when ___.

Veo multiplicación repetida cuando ___.

That situation could be written with a ___ because ___.

Esa situación se podría escribir con una ___ porque ___.

Word bank: **base** **exponent** **power** **evaluate** **growth**

Listen for: *Students give a real doubling or repeated-multiplication example and connect it to a base and exponent.*

Level 2

Compare 3 to the 4th power and 4 to the 3rd power. Are they equal? Evaluate both and use base and exponent to justify your answer.

- 3 to the 4th equals ___ and 4 to the 3rd equals ___, so they are ___.
- Switching the base and exponent changes the value because ___.

Guided Examples

Example 1

What is the value of 4^3?

Solution: $4^3 = 4 \times 4 \times 4 = 64$.

Answer: A. 64

Example 2

What is the value of 6^2?

Solution: $6^2 = 6 \times 6 = 36$.

Answer: A. 36

Example 3

Which expression shows 5^3 as repeated multiplication?

Solution: 5^3 means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.

Answer: A. $5 \times 5 \times 5$

Write About the Math

The Writing Revolution

I can explain my work using the words exponent, base, power, and evaluate.

1. Kernel Sentence

subject + verb

Model: Exponent is a small number that tells how many times to multiply the number by itself.
Exponente es un número pequeño que dice cuántas veces multiplicar el número por sí mismo.

Write a kernel sentence about exponent. Use a subject and a verb.
Escribe una oración base sobre exponente. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Exponent matters in math
Exponente importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Exponent matters in math because ____.
Exponente importa en matemáticas porque ____.

___

but
pero

Exponent matters in math, but ____.
Exponente importa en matemáticas, pero ____.

___

so
entonces

Exponent matters in math, so ____.
Exponente importa en matemáticas, entonces ____.

___

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about exponent.
Di un hecho verdadero sobre exponent.

Exponent ____.

Question
Pregunta

Ask a question about exponent.
Haz una pregunta sobre exponent.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about exponent.
Muestra entusiasmo sobre exponent.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with exponent.
Dile a un compañero qué hacer con exponent.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

Each number is the one before it times ____, so it is repeated ____.
Cada número es el anterior por ____, así que es ____ repetida.

2 times 2 times 2 can be written as 2 to the ____ power.
2 por 2 por 2 se puede escribir como 2 a la ____ potencia.

The base is ____ and it is the number being ____.
La base es ____ y es el número que se está ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which expression shows 5^3 as repeated multiplication?

A. $5 \times 5 \times 5$

B. 5×3

C. $5 + 5 + 5$

D. $3 \times 3 \times 3 \times 3 \times 3$

Show your work:

2. Evaluate: $3 \times (4 + 2)^2$

A. 108

B. 54

C. 36

D. 144

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

Sentence starter: After 6 hours there are ____ bacteria because 2^6 = ____. If it grew by adding 2 each hour instead, there would only be ____ bacteria. Exponential growth is faster because ____.

Show your work:

Reflect — Exit Ticket

What is the value of 3^4?

A. 81

B. 12

C. 34

D. 64

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $5 \times 5 \times 5$ *— 5^3 means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.*
2. **Try It 2:** A. 108 *— $4 + 2 = 6$. $6^2 = 36$. $3 \times 36 = 108$.*
3. **Exit Ticket:** A. 81 *— $3^4 = 3 \times 3 \times 3 \times 3 = 81$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Exponent is a small number that tells how many times to multiply the number by itself.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).